GINHAWA · AIR-QUALITY STATION

# Station Build Manual

Step-by-step instructions to build one Ginhawa monitoring station from the parts the team bought: ESP32, SEN55, A7670SA 4G + GPS, three 18650 cells and a 6 W solar panel. Written for first-time builders. Do the stages in order and do every **Check** before moving on.

**18** stages  **15** figures  about **2–3 days** of work

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STAGE 0 read first

# Before you start

Read this stage once from top to bottom. It covers the four skills the whole build needs: staying safe, using a multimeter, soldering, and preparing wires.

## The station in one picture

You are building a box that measures the air, runs on a battery, recharges itself from the sun, and sends readings to the Ginhawa app over Wi-Fi or 4G. The picture below is the whole station. Every stage in this manual builds one part of it.

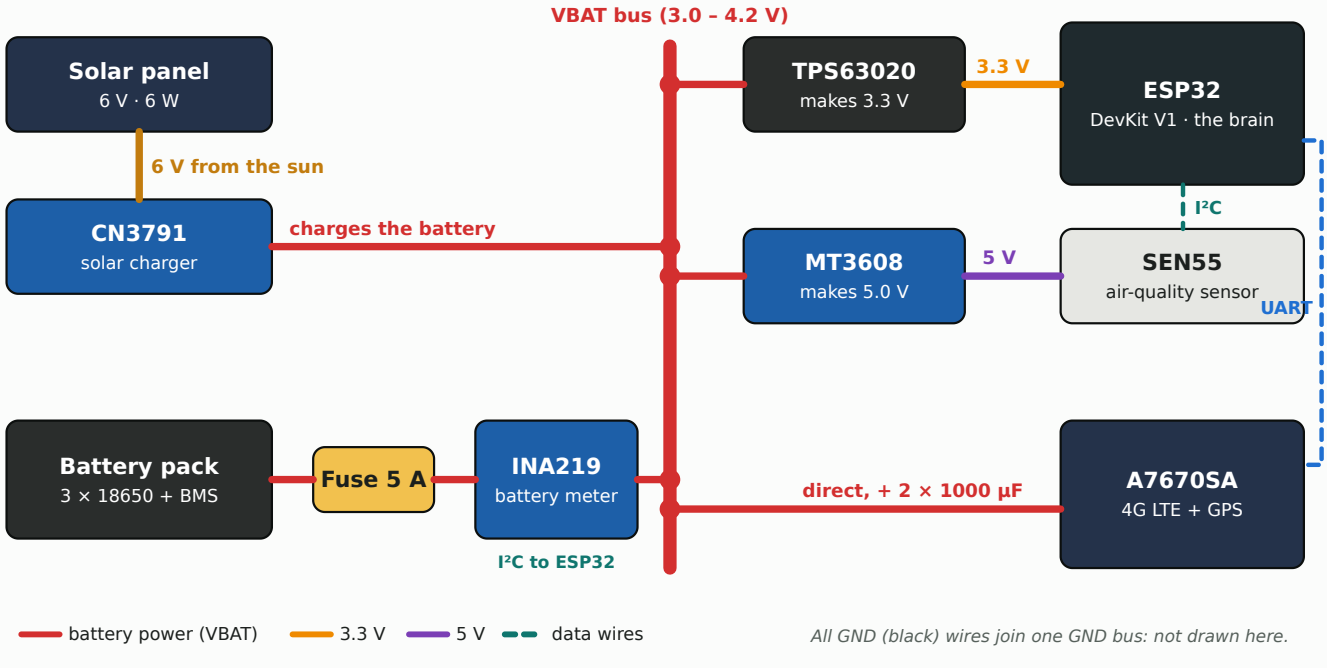



**Figure 1.** How power and data move through the station. The battery feeds one **VBAT bus**; every module takes its power from that bus. The solar panel refills the battery through the CN3791.

## Safety rules

**LITHIUM BATTERIES CAN BURN IF SHORTED**

- Never let a bare red and black battery wire touch each other. Wrap loose battery wire ends in electrical tape until they are soldered.
- Never solder directly onto a battery cell. The cells only go in the holder.
- Insert the cells **last**, only when the stage tells you to. Take them out before you solder anything on the power path.
- Work on a wooden or ceramic surface, never on a bed or carpet. Keep a metal tray or sand nearby for a cell that gets hot.
- If a cell gets hot, swells or smells sweet: take it out with pliers, put it outside on concrete, and do not use it again.

**SOLDERING SAFETY**

- The iron tip is about 350 °C. Always put it back in its stand. Unplug it when you leave the table.
- Solder smoke irritates: open a window or use a fan blowing the smoke away from your face.
- Wear glasses when you cut component legs. Wash your hands after soldering.

## How to use the multimeter

You only need two settings. Use them at every **Check** box in this manual.

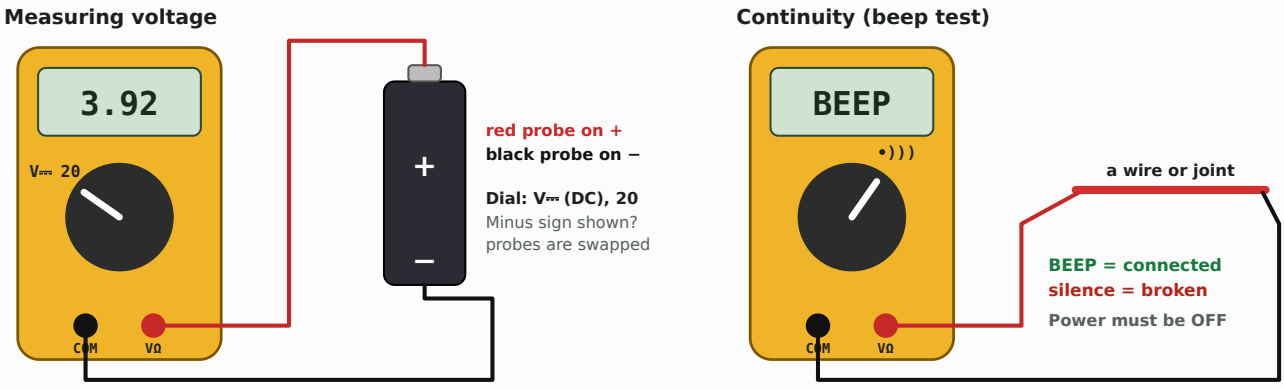



**Figure 2.** The two multimeter settings you need. Black probe always goes in **COM**, red in **VΩ**.

- **Voltage (V⎓, DC, range 20):** the circuit is ON. Touch red to + and black to − (or GND). Read the number.
- **Continuity (the speaker or •))) symbol):** the circuit is OFF (cells out, fuse out). Touch the probes to two points. A beep means they are connected.

## How to solder (5 steps)

1. Heat the iron to about 350 °C. Wipe the tip on a damp sponge or brass wool, then melt a little solder on the tip. A shiny tip conducts heat well.

2. Touch the tip so it heats **both** the pad and the wire or pin at the same time. Count 1–2 seconds.

3. Feed solder into the joint (on the opposite side from the tip, not onto the tip). Stop when the pad is covered: about 1 second.

4. Remove the solder first, then the iron. Do not move the part for 3 seconds while it cools.

5. Look at the joint and compare it with Figure 3. Cut extra leg length with the side cutter.

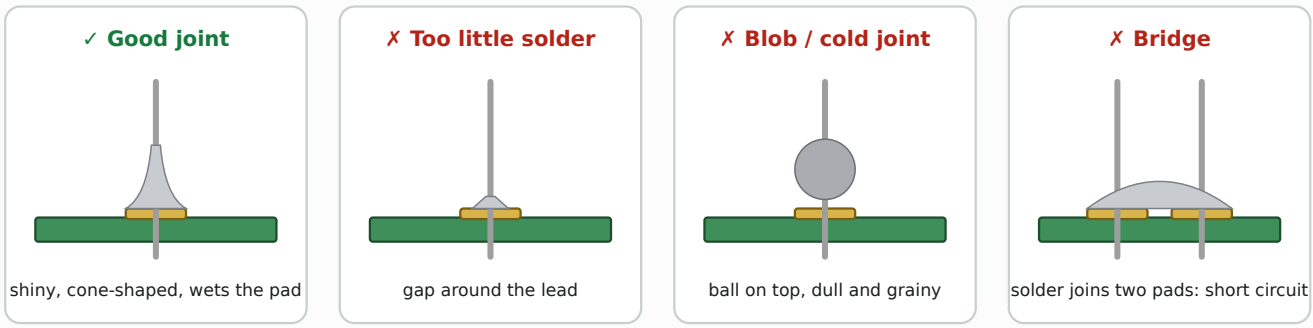



**Figure 3.** What solder joints should look like. Fix any ✗ joint: reheat it and add or remove solder (use the tip to drag extra away).

## How to prepare a wire

1. Cut the wire a little longer than needed (about 2 cm extra).

2. Strip 5 mm of insulation off each end (wire stripper or carefully with the side cutter).

3. Twist the strands together, then **tin** the end: heat it and let it soak up a little solder. A tinned end is easy to solder and never frays.

4 **Slide heat-shrink onto the wire before soldering the second end.** After soldering, slide it over the joint and shrink it with a hair dryer on high or a heat gun.

5 Mark both ends of each signal wire with the same colour of heat-shrink (yellow = SDA, green = SCL, blue/cyan = modem). You only have red and black wire, so the tags tell the wires apart.

**COLOURS IN THIS MANUAL**

**Red** = battery +, **black** = GND (ground), **orange** = 3.3 V, **purple** = 5 V. Use red wire for every + and black for every GND. For signal wires use any wire with coloured tags.

STAGE 1 30 min

# Check every part when it arrives

Most sellers give 7 days for returns. Test each part the day it arrives, before you solder anything to it.

| PART | WHAT TO CHECK | RETURN IT IF |
|---|---|---|
| A7670SA breakout | The metal shield says **A7670SA-FASE**. Two antenna connectors (MAIN and GNSS). Both antennas in the bag. | Says LASE or LASC (no GPS) |
| Beston 18650 × 4 | Weigh each one: about **46 g**. Measure voltage: all within 0.05 V of each other. | Under 40 g, or below 2.5 V |
| 3-cell holder | All springs on one side. With cells inside, the leads read 3.6–4.2 V (Figure 4). | Reads about 11–12 V (series) |
| Solar panel | In direct sun, the red and black wires read **6–7.5 V**. | Below 5 V in full sun |
| CN3791 | The listing option was **6V**. In sun, with the panel connected and no battery, the BAT side reads about 4.2 V. | BAT side reads 0 V |
| TPS63020 | The option was **3.3V**. Test in Stage 7. | Output is not 3.25–3.35 V |
| BMS | Printed **3.7V** (not 3.2V). Pads B+, B−, P− (and maybe P+). | Says 3.2V / LiFePO4 |
| INA219 | Blue board, **R100** on the big resistor, a green screw terminal and a pin strip in the bag. | Missing parts |
| JST GH cable | 6 pins, 1.25 mm. The plug fits the SEN55 connector gently (do not force it). | Wrong size |

**KEEP A PARTS BOX**

Put the small parts (resistors, capacitors, screws, fuses) in a labelled container. Losing a 4.7 kΩ resistor costs a day.

STAGE 2 15 min

# Prepare the battery cells

Cells connected in parallel must start at the same voltage. Otherwise the full cell pushes a large current into the empty one the moment they touch.

1. Measure each of the 4 Beston cells (multimeter on V⎓ 20). Write the voltage on a piece of masking tape stuck to each cell.

2. Pick the **3 cells** whose voltages are closest together. The 4th is your spare: store it in a dry box, away from metal.

3. If the 3 cells are all within **0.05 V** of each other, go to Stage 3.

4. If not: build Stages 3 to 6 without the cells. Then put only **one** cell in the holder and let the solar panel charge it until it reads 4.15–4.20 V. Repeat for each cell. When all 3 read the same, put them in together. (The CN3791 stops charging at 4.2 V by itself.)

**CHECK**

All 3 cells within 0.05 V of each other, for example 3.71 V, 3.72 V and 3.74 V.

STAGE 3 45 min

# Build the battery pack

Connect the holder to the BMS. The BMS cuts the power if the battery is over-charged, over-drained or shorted.

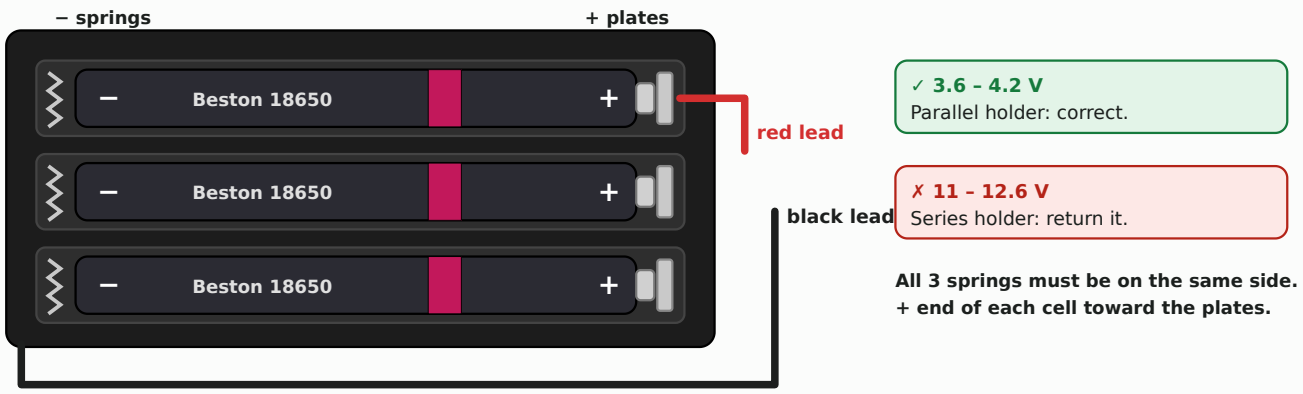



**Figure 4.** Cells in the 3-cell **parallel** holder. Every cell faces the same way: + end on the metal plates, − end on the springs.

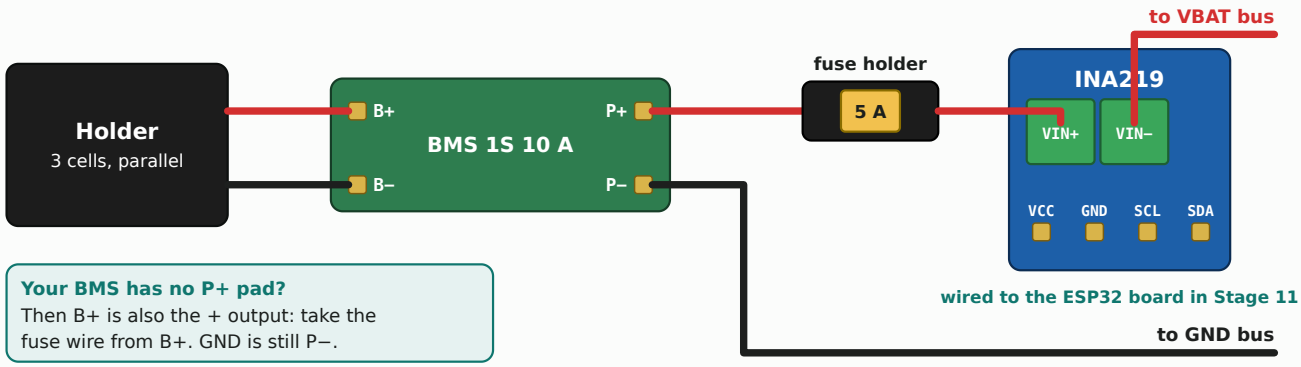



**Figure 5.** Battery → BMS → fuse → INA219. The thick red path carries all the station's current. **P−** becomes the station's ground (GND).

1 With **no cells in the holder**, cut the holder's thin leads to 5 cm. Solder an 18 AWG red wire (10 cm) to the red lead and an 18 AWG black wire (10 cm) to the black lead. Cover both joints with heat-shrink.

2 Look at the BMS. Find the pads printed **B+**, **B−**, **P−** and maybe **P+**. Cut its nickel strips to about 5 mm (you will solder to them, not weld).

3 Solder the holder's red wire to **B+** and the holder's black wire to **B−**.

4 Solder an 18 AWG red wire (15 cm) to **P+** (or to B+ if there is no P+). This goes to the fuse in Stage 5.

5 Solder an 18 AWG black wire (15 cm) to **P−**. This is the station's main GND wire. Wrap both loose ends in tape.

6 Put the 3 matched cells in the holder: flat + end against the metal plates, − end against the springs.

**CHECK**

| MEASURE BETWEEN | MULTIMETER ON | EXPECTED |
|---|---|---|
| Holder red and black leads | V⎓ 20 | **3.6 – 4.2 V** |
| BMS P+ (or B+) and P− | V⎓ 20 | **same as above** |

P+ to P− reads 0 V? The BMS is asleep. It wakes when the charger is connected in Stage 6.

**NEVER CONNECT ANYTHING TO B−**

All grounds go to **P−**. A ground wire on B− bypasses the BMS protection.

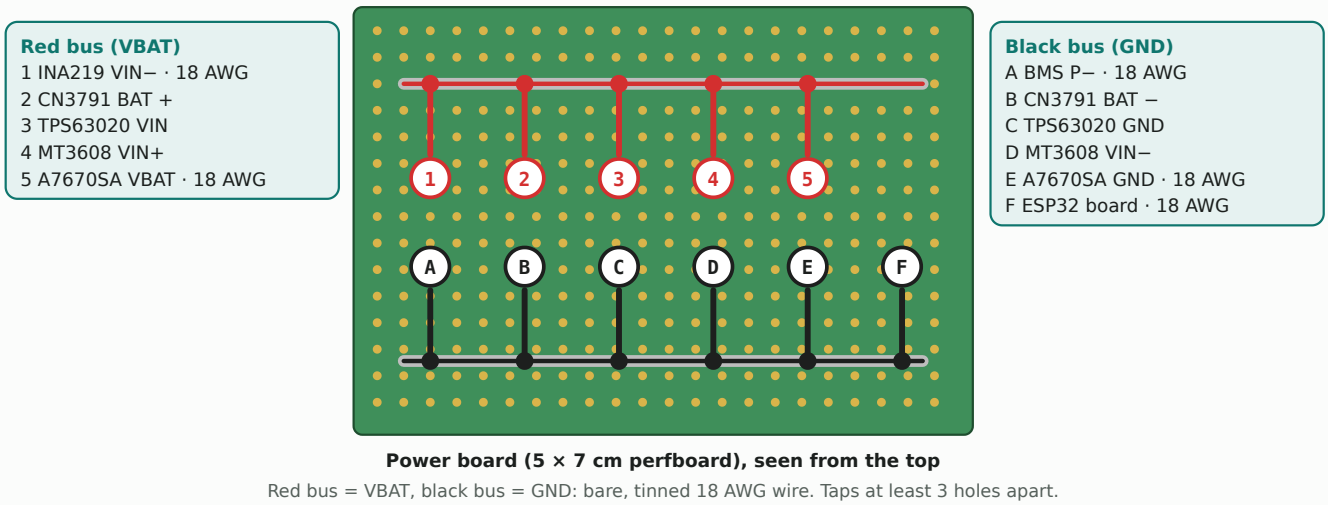
**TAKE THE CELLS OUT NOW**

Remove the cells until Stage 6. Stages 4 and 5 involve soldering on the power path.

STAGE 4 30 min

# Make the power board

A small perfboard with two bus wires: red for VBAT, black for GND. All modules take their power from it.




**Power board (5 × 7 cm perfboard), seen from the top**

Red bus = VBAT, black bus = GND: bare, tinned 18 AWG wire. Taps at least 3 holes apart.

**Figure 6.** The power board is two long bus wires. Everything that needs battery power solders to the red bus (taps 1–5); every ground solders to the black bus (taps A–F).

1. Take the **5 × 7 cm** perfboard.

2. Cut two pieces of 18 AWG wire, each 7 cm. Strip the **whole** insulation off and tin them: these are the bus wires.

3. Lay the first bare wire along the top row of holes and solder it every 4–5 holes. This is the **VBAT bus**. Mark it with red tape or a red marker.

4. Lay the second along the bottom row and solder it the same way. This is the **GND bus**. Mark it black.

5. Each wire that connects later goes through a hole next to its bus and is soldered to the bus wire. Space the joints at least 3 holes apart.

**CHECK (NO POWER CONNECTED)**

| MEASURE BETWEEN | MULTIMETER ON | EXPECTED |
|---|---|---|
| VBAT bus and GND bus | continuity | **no beep (not shorted)** |
| Both ends of the VBAT bus | continuity | **beep** |

STAGE 5 40 min

# Fuse and battery meter (INA219)

The fuse protects the wiring from a short circuit. The INA219 measures the battery voltage and current for the app.

Follow the right half of Figure 5.

1. Solder the INA219's green screw terminal onto the board (2 pins, labelled VIN+ and VIN−). Then solder the 6-pin header strip on its other edge.
2. Cut one wire of the inline fuse holder in the middle. You now have two ends: strip and tin both.
3. Join the BMS **P+** wire to one fuse-holder end: twist, solder, cover with heat-shrink.
4. Screw the other fuse-holder end into the INA219 **VIN+** terminal. Tighten firmly.
5. Screw a new 18 AWG red wire (10 cm) into **VIN−**. Solder its other end to the **VBAT bus**.
6. Solder the BMS **P−** black wire to the **GND bus**.
7. Leave the fuse **out** of the holder. From now on, the fuse is the station's ON/OFF switch.

**CHECK: CELLS IN, THEN INSERT THE 5 A FUSE**

| MEASURE BETWEEN | MULTIMETER ON | EXPECTED |
|---|---|---|
| VBAT bus and GND bus | V⎓ 20 | **3.6 – 4.2 V** |

Pull the fuse out again before the next stage.

**WHICH FUSE**

Use the **5 A** fuse (usually tan or light brown, with 5 printed on top). A 7.5 A fuse also works.

STAGE 6 40 min

# Solar panel and charger (CN3791)

The charger takes 6 V from the panel and refills the battery, stopping at 4.2 V.

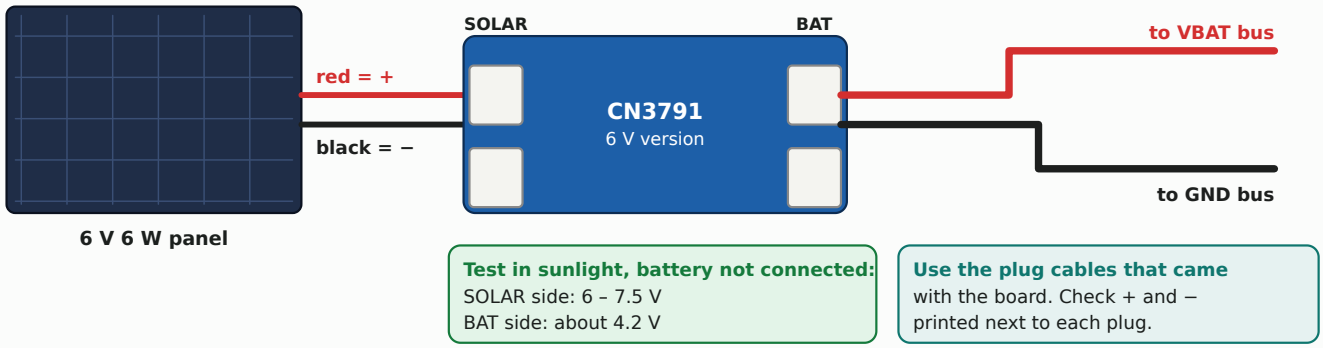



**Figure 7.** Solar panel → CN3791 → battery bus. Either SOLAR plug works (they are connected together), and the same for BAT.

1. Cut the plug off the solar panel cable, leaving as much cable as possible. Strip and tin the two wires.

2. In sunlight, measure the two panel wires to find + (the meter shows a positive number when red probe is on +). Mark + with red tape.

3. Take one of the plug cables that came with the CN3791. Plug it into a **SOLAR** socket and check which wire matches the + printed on the board.

4. Solder the panel + to that cable's + wire and the panel − to its − wire. Cover with heat-shrink, then wrap with electrical tape (this joint will be near the outside).

5. Plug a second cable into a **BAT** socket. Solder its + wire to the **VBAT bus** and its − wire to the **GND bus**.

6. Insert the cells and the fuse. Put the panel in the sun.

**CHECK**

| MEASURE BETWEEN | MULTIMETER ON | EXPECTED |
|---|---|---|
| CN3791 SOLAR + and − | V⎓ 20 | **5 – 7.5 V in sun** |
| VBAT bus and GND bus | V⎓ 20 | **rises slowly while charging, max 4.2 V** |

The CN3791 has small LEDs: one lights while charging, the other when the battery is full.

STAGE 7 20 min

# 3.3 V converter (TPS63020)

Makes a steady 3.3 V for the ESP32 and the INA219, whether the battery is full (4.2 V) or low (3.0 V).

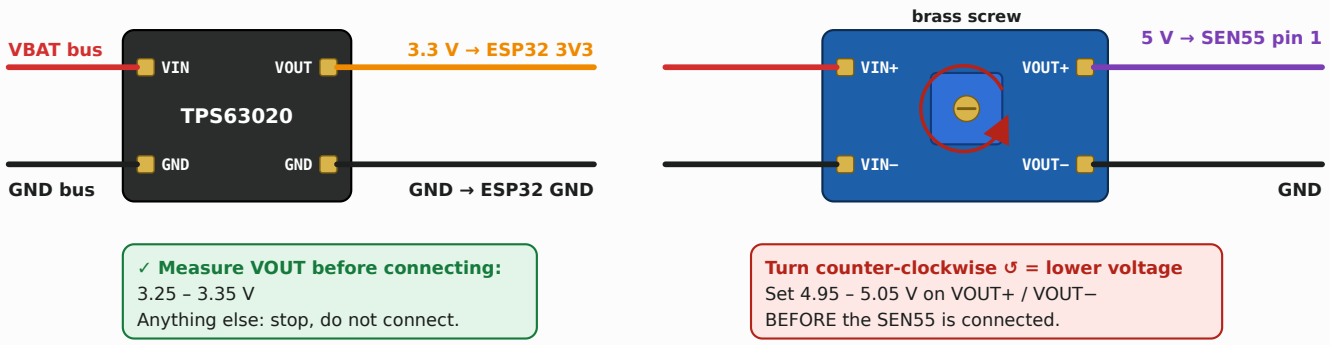



**Figure 8.** The two voltage converters. Both take power from the VBAT bus. Measure their outputs before anything is connected to them.

1. Pull the fuse out (power off).

2. Solder a 22 AWG red wire from TPS63020 **VIN** to the VBAT bus, and a black wire from its input **GND** to the GND bus.

3. Solder a 22 AWG red wire (20 cm, tag it orange) to **VOUT** and a black wire (20 cm) to the output **GND**. Leave their other ends free and taped for now.

4. Insert the fuse and measure VOUT to GND.

**CHECK**

| MEASURE BETWEEN | MULTIMETER ON | EXPECTED |
|---|---|---|
| TPS63020 VOUT and GND | V⎓ 20 | **3.25 – 3.35 V** |

Reads 5 V? You received the 5 V version: do not connect it to the ESP32. Pull the fuse and contact the seller.

STAGE 8 20 min

# 5 V converter (MT3608)

Makes 5 V for the SEN55 sensor. It arrives set to a random voltage, often 12 V or more, which would destroy the sensor.

1. Pull the fuse out. Solder MT3608 **VIN+** to the VBAT bus and **VIN−** to the GND bus (22 AWG).
2. Solder a 22 AWG red wire (20 cm, tag it purple) to **VOUT+** and a black wire (20 cm) to **VOUT−**. Tape the free ends.
3. Insert the fuse. Put the multimeter on VOUT+ and VOUT−.
4. With a small screwdriver, turn the brass screw on the blue part **counter-clockwise** (↺). The voltage drops. It can take 10–20 turns before it starts to move.
5. Stop at **5.0 V** (between 4.95 and 5.05 V). If you go too low, turn clockwise a little.
6. Put a drop of nail polish or hot glue on the screw so it cannot turn by vibration.

**CHECK**

| MEASURE BETWEEN | MULTIMETER ON | EXPECTED |
|---|---|---|
| MT3608 VOUT+ and VOUT− | V⎓ 20 | **4.95 – 5.05 V** |

**DO NOT SKIP THIS STEP**

The SEN55 accepts 4.5–5.5 V. More than that destroys a sensor that costs several thousand pesos.

STAGE 9 1 – 2 h

# Make the ESP32 board

The ESP32 plugs into sockets on the larger perfboard, so you can take it out to program it or replace it.

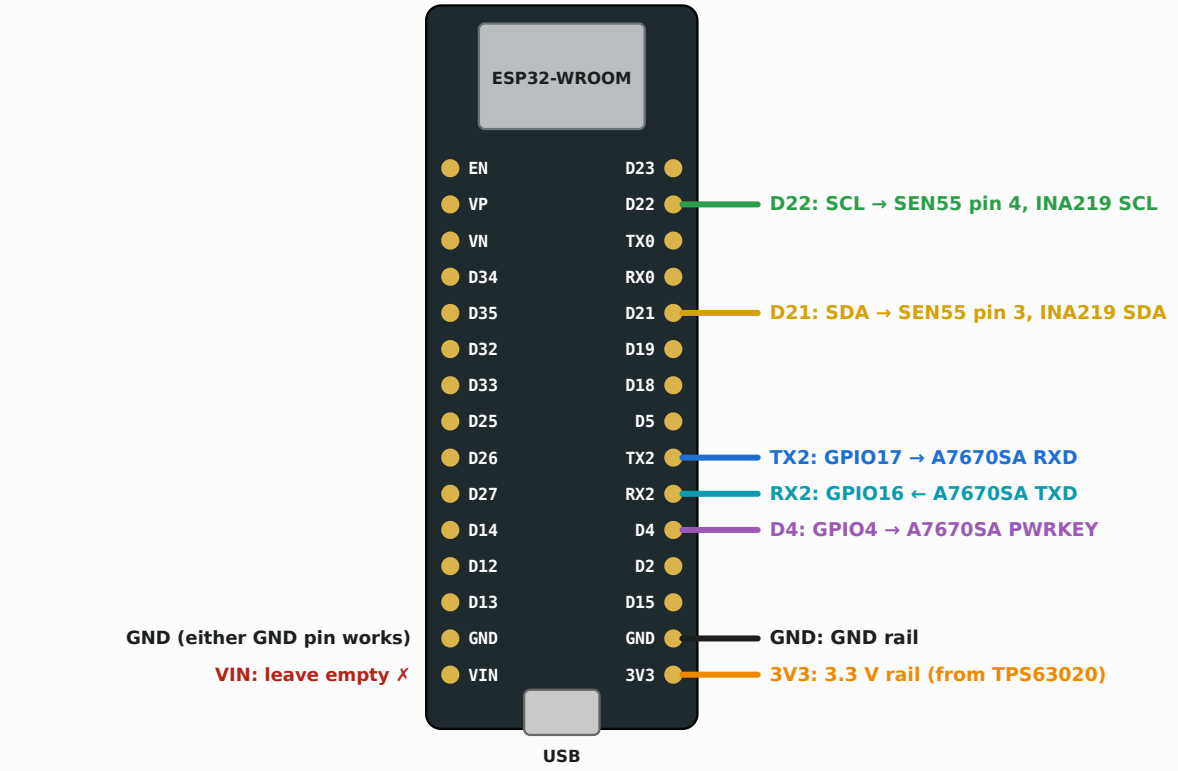



**Figure 9.** The ESP32 pins this project uses, all on one side of a 30-pin DevKit V1. **TX2 = GPIO17** and **RX2 = GPIO16** on the board's labels.

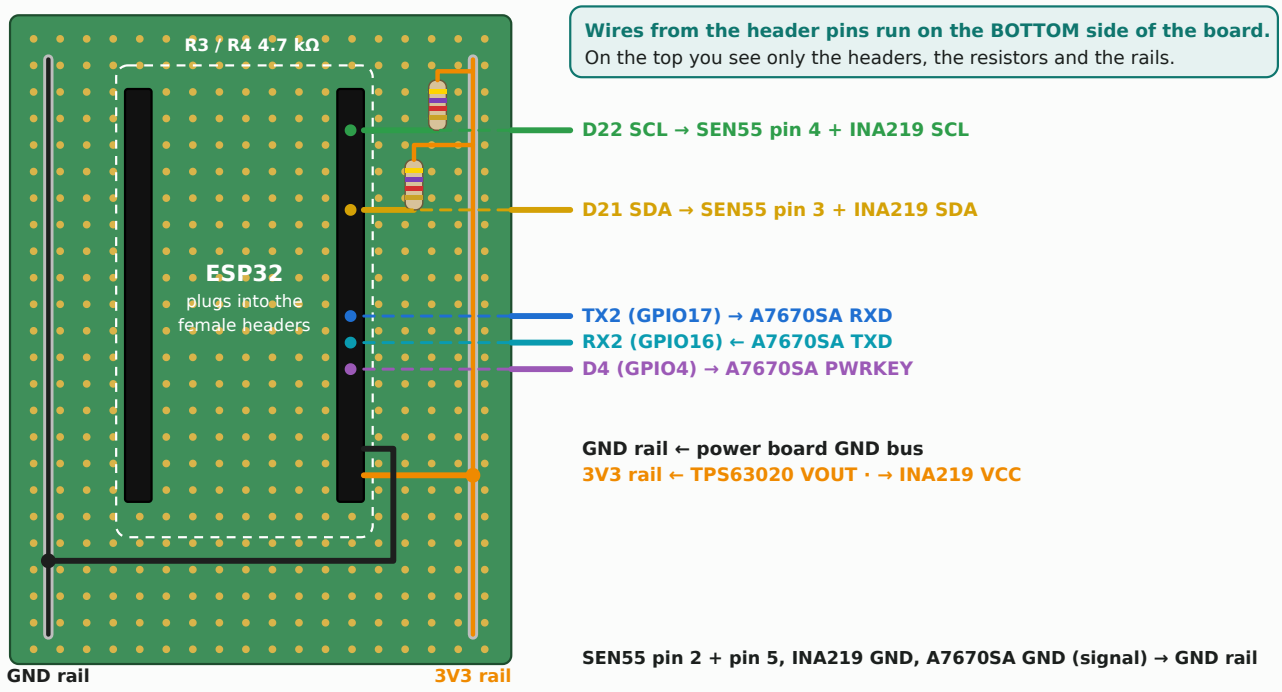



**Figure 10.** The ESP32 board (7 × 9 cm). Two rails (GND and 3V3), two pull-up resistors, and solder points for the sensor and modem wires.

1. Take the 7 × 9 cm perfboard. Count the pins on one side of your ESP32 (15 on a 30-pin board). Cut two pieces of that length from the female header strip (pull out the metal pin where you want to cut, then cut through the empty slot).

2. **Plug the two headers onto the ESP32's pins first**, then set the ESP32 with the headers onto the perfboard. This guarantees the right spacing.

3. Solder one pin at each end of each header, check everything sits straight, then solder all the remaining pins. Lift the ESP32 off.

4. Run a bare tinned wire along the left edge (**GND rail**) and one along the right edge (**3V3 rail**), soldered every few holes, like the power board.

5. On the bottom side, connect the header pin under ESP32 **3V3** to the 3V3 rail, and the pin under **GND** to the GND rail.

6. Solder R3 (4.7 kΩ) between the **D21** header pin and the 3V3 rail, and R4 (4.7 kΩ) between **D22** and the 3V3 rail. Resistors have no polarity.

7 Connect the TPS63020 output wires: orange-tagged **VOUT** → 3V3 rail, its black GND → GND rail.

8 Connect a thick black wire from the ESP32 board GND rail to the power board **GND bus**.

**CHECK: ESP32 NOT PLUGGED IN, FUSE IN**

| MEASURE BETWEEN | MULTIMETER ON | EXPECTED |
|---|---|---|
| 3V3 rail and GND rail | V⎓ 20 | **3.25 – 3.35 V** |
| Header pin under VIN and anything | continuity | **no beep: VIN stays empty** |

STAGE 10 30 min

# Set up the computer and program the ESP32

Install the free Arduino IDE once. Program the ESP32 on its own, out of the board, powered by the USB cable.

1. Download and install **Arduino IDE 2** from arduino.cc/en/software.

2. Open **File → Preferences**. In "Additional boards manager URLs" paste: `https://espressif.github.io/arduino-esp32/package_esp32_index.json`

3. Open **Tools → Board → Boards Manager**, search **esp32**, install **esp32 by Espressif Systems**.

4. Open **Tools → Manage Libraries** and install: **Sensirion I2C SEN5X** (accept "Sensirion Core" too), **Adafruit INA219** (accept "Adafruit BusIO"), and **ArduinoJson** (version 7).

5. Plug the ESP32 (out of the board) into the computer with a USB **data** cable. Select **Tools → Board → esp32 → ESP32 Dev Module** and **Tools → Port → COM…** (the new one that appeared).

6. No COM port appears? Install the USB driver printed on the small chip near the USB connector: **CP2102** (Silicon Labs CP210x driver) or **CH340** (WCH CH340 driver). Or try another cable: many cables only charge.

7. Open **File → Examples → 01.Basics → Blink**, add `#define LED_BUILTIN 2` at the top, and click **Upload** (→ arrow).

8. If it stays at "Connecting.....", hold the **BOOT** button on the ESP32 until the upload starts.

**CHECK**

The small blue LED on the ESP32 blinks once per second. Your computer, cable and ESP32 all work.

STAGE 11 1 h

# Connect the SEN55 and INA219, then test

Both talk to the ESP32 over two shared wires called I²C: SDA (data) and SCL (clock).

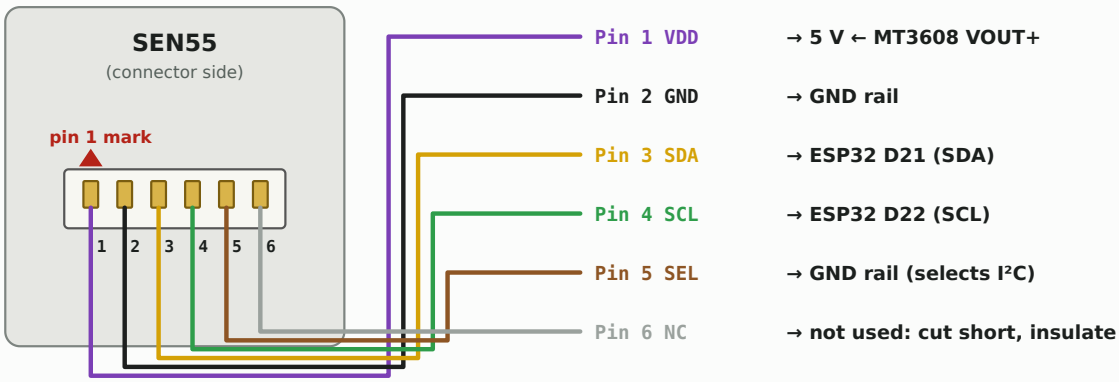



**Figure 11.** SEN55 pin-out and where each wire goes. Find **pin 1** from the mark printed next to the sensor's connector, then count. Go by pin number, never by wire colour.

1. Cut one double-ended JST GH 6-pin cable in the middle. Plug one half into the SEN55. Do not cut the spare half.

2. Use continuity mode to find which bare wire goes to **pin 1** of the plug (look for the mark next to the SEN55 connector). Write down the colour of each pin 1–6.

3. Pull the fuse out. Solder: pin 1 → MT3608 **VOUT+** (the purple-tagged wire); pin 2 → GND rail; pin 3 → **D21** header pin; pin 4 → **D22** header pin; pin 5 → GND rail. Cut pin 6 short and cover it with heat-shrink.

4. Join the MT3608 **VOUT−** black wire to the ESP32 board GND rail.

5. INA219 header pins: **VCC** → 3V3 rail, **GND** → GND rail, **SDA** → D21, **SCL** → D22 (four 22 AWG wires).

6. Plug the ESP32 into its headers, USB end toward the board edge. Insert the fuse, then connect the USB cable.

7 Copy the test sketch below into a new Arduino sketch, upload it, then open **Tools → Serial Monitor** and set it to **115200 baud**.

**Test 1: I²C scanner**

```
// Ginhawa test 1: finds devices on the I2C bus (SDA = D21, SCL = D22)
#include <Wire.h>

void setup() {
  Serial.begin(115200);
  delay(1000);
  Wire.begin();  // SDA 21, SCL 22
}

void loop() {
  Serial.println("Scanning...");
  int found = 0;
  for (byte addr = 1; addr < 127; addr++) {
    Wire.beginTransmission(addr);
    if (Wire.endTransmission() == 0) {
      Serial.printf("  found 0x%02X", addr);
      if (addr == 0x40) Serial.print("  <- INA219 (battery meter)");
      if (addr == 0x69) Serial.print("  <- SEN55 (air sensor)");
      Serial.println();
      found++;
    }
  }
  if (found == 0) Serial.println("  nothing found: check SDA/SCL, power and GND");
  Serial.println();
  delay(3000);
}
```

**CHECK**

The Serial Monitor shows `found 0x40 <- INA219` and `found 0x69 <- SEN55`. You may also hear the SEN55 fan start briefly.

**USB AND BATTERY TOGETHER**

Having the USB cable and the fuse in at the same time is fine for these short tests. When the station is finished, unplug USB for good: the station runs only from its battery.

STAGE 12 1 – 2 h

# Connect the 4G modem (A7670SA) and test

The modem sends readings over 4G when there is no Wi-Fi, and its built-in GPS gives the station's location.

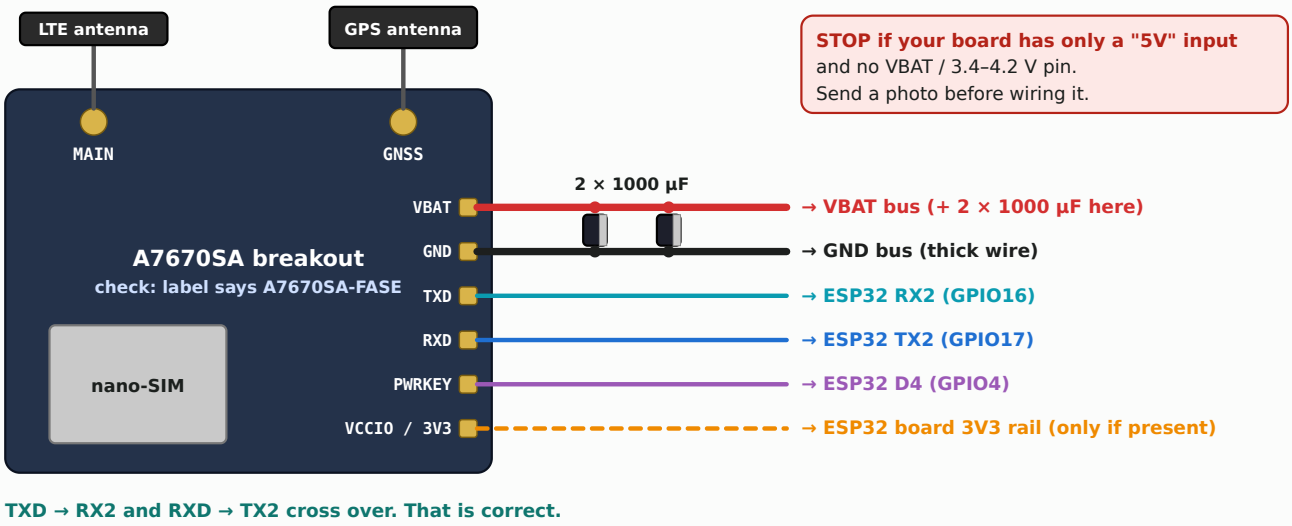



**Figure 12.** A7670SA connections. Pin names differ between breakout makers: PWRKEY may be printed as **PWR**, **PEN** or **K**; VBAT may be **VCC 3.4–4.2 V** or **BAT**.

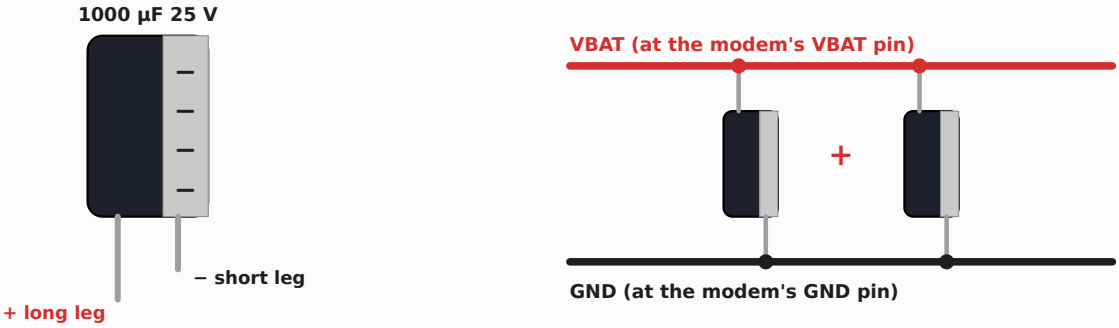



**Figure 13.** Capacitor polarity. The side with the stripe is **−**. Two capacitors side by side, both + to VBAT and both − to GND.

1 Compare your board with Figure 12. Find these pins: **VBAT** (or VCC 3.4–4.2 V), **GND**, **TXD**, **RXD**, **PWRKEY** (or PWR, PEN, K). Some boards also have **VCCIO** or **3V3** for the signal level.

2 Board has only a 5 V input and no VBAT pin? **Stop here** and send a clear photo of both sides of the board before wiring.

3 Pull the fuse out. Solder an 18 AWG red wire from the VBAT bus to the modem **VBAT**, and an 18 AWG black wire from the GND bus to the modem **GND**. Keep these two wires short (under 15 cm).

4 Solder the two 1000 µF capacitors right at the modem's VBAT and GND pins: long leg (+) to VBAT, striped side (−) to GND (Figure 13).

5 Signal wires (22 AWG): modem **TXD** → ESP32 **RX2** pin; modem **RXD** → ESP32 **TX2** pin; modem **PWRKEY** → ESP32 **D4**. If the board has VCCIO/3V3, connect it to the 3V3 rail.

6 Connect the antennas: the LTE antenna to **MAIN** (sometimes "LTE"), the GPS antenna to **GNSS**. Press the small round plugs straight down until they click.

7 Insert a registered **nano-SIM** with load (cut corner as shown on the slot). Insert the fuse.

8 Upload Test 2 below. In the Serial Monitor choose **115200 baud** and **Both NL & CR**, wait 15 seconds, then type the commands in the table one at a time.

**Test 2: modem terminal**

```
// Ginhawa test 2: talk to the A7670SA. Type AT commands in the Serial Monitor.
// Serial Monitor: 115200 baud, "Both NL & CR".
#define MODEM_RX 16   // ESP32 RX2 <- modem TXD
#define MODEM_TX 17   // ESP32 TX2 -> modem RXD
#define PWRKEY    4   // ESP32 D4  -> modem PWRKEY

void setup() {
  Serial.begin(115200);
  Serial2.begin(115200, SERIAL_8N1, MODEM_RX, MODEM_TX);
  pinMode(PWRKEY, OUTPUT);
  Serial.println("Pressing the modem's power key for 1 second...");
  digitalWrite(PWRKEY, HIGH);
  delay(1000);
  digitalWrite(PWRKEY, LOW);
  Serial.println("Wait about 15 seconds, then type: AT");
}

void loop() {
  while (Serial.available())  Serial2.write(Serial.read());   // you -> modem
  while (Serial2.available()) Serial.write(Serial2.read());   // modem -> you
}
```

| TYPE | GOOD ANSWER | MEANING |
|---|---|---|
| AT | OK | The modem is on and talking |
| AT+SIMCOMATI | Model: A7670SA… | Shows the model. Look for FASE in the details. |
| AT+CPIN? | +CPIN: READY | The SIM card is detected |
| AT+CSQ | +CSQ: 15,99 | Signal: first number 10–31 is usable, 99 means none |
| AT+CEREG? | +CEREG: 0,1 | Registered on 4G (0,5 = roaming, also fine) |
| AT+CGNSSPWR=1 | +CGNSSPWR: READY! | GPS switched on |
| AT+CGPSINFO | +CGPSINFO: 1429.12,N,… | A position. Empty commas = no fix yet: go outside, wait 1–5 min, repeat |

**NO ANSWER TO AT?**

- Swap the TXD and RXD wires (some boards label them from the ESP32's side).
- Check the board's power LED. Press its power button if it has one.
- Your board's PWRKEY may need the opposite signal: tell us what the board's LEDs do and we will adjust the firmware.

STAGE 13 30 min

# Upload the Ginhawa firmware

This is the real program. It reads the sensor every second, uploads once a minute, and adds the station to the app by itself.

1. Download the project from GitHub (green **Code → Download ZIP**) or use your `C:\Users\peter\ginhawa` folder. Open `firmware/ginhawa_node/ginhawa_node.ino` in Arduino IDE.

2. In the same folder, copy `config.example.h` and rename the copy to `config.h`. Open it from the tab at the top of the Arduino IDE.

3. Fill in the five lines below. Keep the quote marks.

4. Find your laptop's address: on Windows open PowerShell, type `ipconfig`, and copy the **IPv4 Address** of your Wi-Fi (for example 192.168.1.20). The station and laptop must be on the same Wi-Fi, and the Ginhawa app must be running (Docker).

5. Click **Upload**. Open the Serial Monitor at 115200 baud.

**config.h: the lines you change**

```
#define WIFI_SSID      "YourWiFiName"
#define WIFI_PASSWORD  "YourWiFiPassword"
#define API_BASE_URL   "https://192.168.1.20"      // your laptop's IP address (see
below)
#define ENROLL_SECRET  "same-as-ENROLL_SECRET-in-.env"
#define LTE_APN        "internet.dito.ph"          // DITO. Smart: "internet"  Globe:
"internet.globe.com.ph"
```

**CHECK: THE SERIAL MONITOR SHOWS, IN ORDER**

1. Chip ID … will enroll on first upload
2. Connecting to YourWiFiName….. connected
3. Ginhawa station ready.
4. Enrolled as "New station (…)"
5. Uploaded 1 reading(s) over Wi-Fi

**LTE NEEDS A PUBLIC SERVER**

Over 4G the station cannot reach a laptop at home. For 4G uploads the app must be online (domain or VPS). Wi-Fi uploads to the laptop work for testing and the defense.

STAGE 14 1 h

# Prepare the enclosure

Drill the box before anything goes inside. Measure twice, drill once.

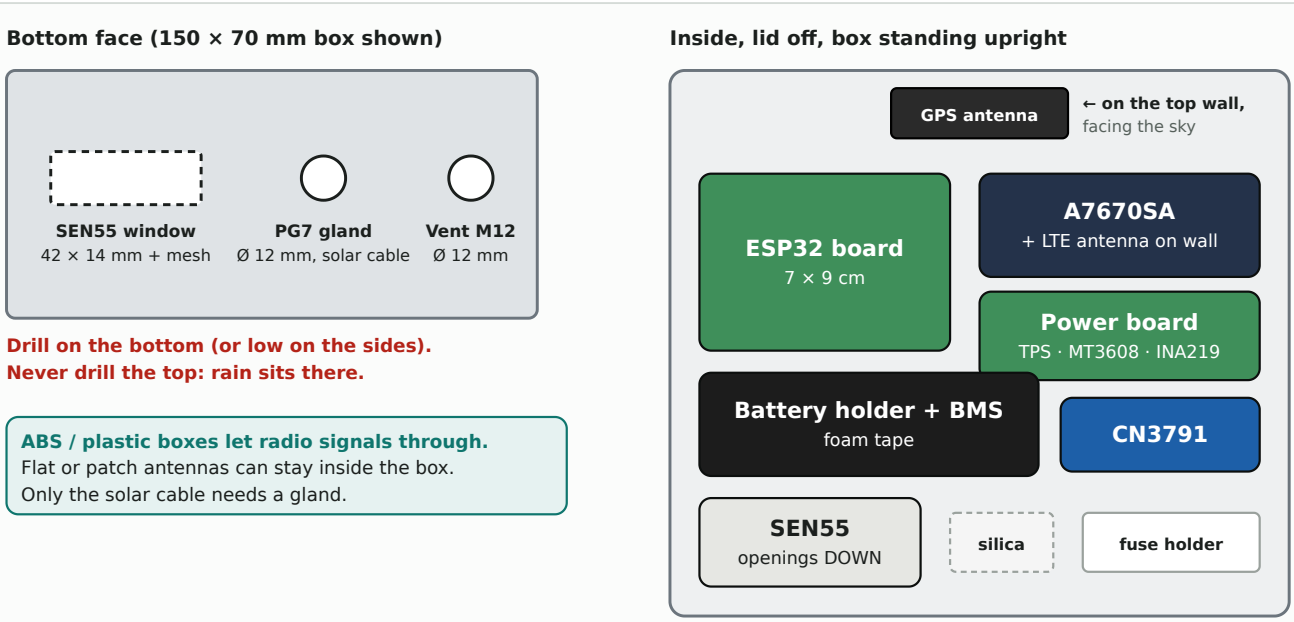



**Figure 14.** Drilling plan and layout inside an IP65 box of about 150 × 100 × 70 mm. Adjust the positions to the box you buy, keeping these rules.

1. Buy an **IP65 ABS junction box**, about 150 × 100 × 70 mm or bigger. Lay all your parts in it first to check they fit with the lid closed.

2. Mark the holes on the **bottom** face with a marker: one 12 mm hole for the PG7 gland (solar cable), one 12 mm hole for the vent, and the SEN55 window (42 × 14 mm).

3. Drill a 3 mm pilot hole first, then the 12 mm bit (or a step bit). Drill slowly so the plastic does not crack.

4. For the window: drill holes in the four corners, then cut between them with a cutter. Smooth the edges with a file.

5. Cut the stainless mesh 10 mm larger than the window on every side. Glue it on the **inside** with hot glue or epoxy all around. No gaps: insects follow air.

6 Install the gland: body through the hole from outside, nut inside, tighten by hand plus a quarter turn. Install the vent the same way, O-ring on the outside.

7 Drill four 3 mm holes in the back wall for the ESP32-board standoffs, matching the board's corner holes.

STAGE 15 1 – 2 h

# Put everything in the box

Fix every part so nothing can move or touch another when the box shakes on a pole.

1. Pull the fuse out.

2. Screw the four M3 standoffs to the back wall with M3 × 8 screws from outside (2 mm Allen key). Fit the ESP32 board on the standoff threads and hold it with M3 nuts.

3. Stick the battery holder (cells out), power board, CN3791 and modem to the box with double-sided foam tape. Put a zip tie or tape strip around the holder so the cells cannot fall out.

4. Mount the SEN55 with its two air openings facing **down**, pressed against the mesh window. Foam tape on its back. Keep it at least 3 cm from the modem and converters.

5. Stick the GPS antenna to the inside of the **top** wall, flat side facing up. Stick the LTE antenna flat on a side wall, away from the GPS antenna.

6. Pass the solar cable through the gland from outside, leave a small drip loop (a U shape) outside, and tighten the gland until the cable cannot be pulled.

7. Bundle loose wires with small zip ties. No bare metal may touch the box, another board or the mesh.

8. Put in the cells, then the fuse. Add 1–2 silica gel packs. Close the lid evenly so the rubber seal presses all around.

**CHECK BEFORE CLOSING**

- Serial Monitor (USB connected briefly) shows "Uploaded … over Wi-Fi".
- Gently shake the box: nothing rattles.
- Unplug USB. Close the lid.

STAGE 16 1 h

# Mount it outside

Place the station where it measures the air people breathe, with sun on the panel.

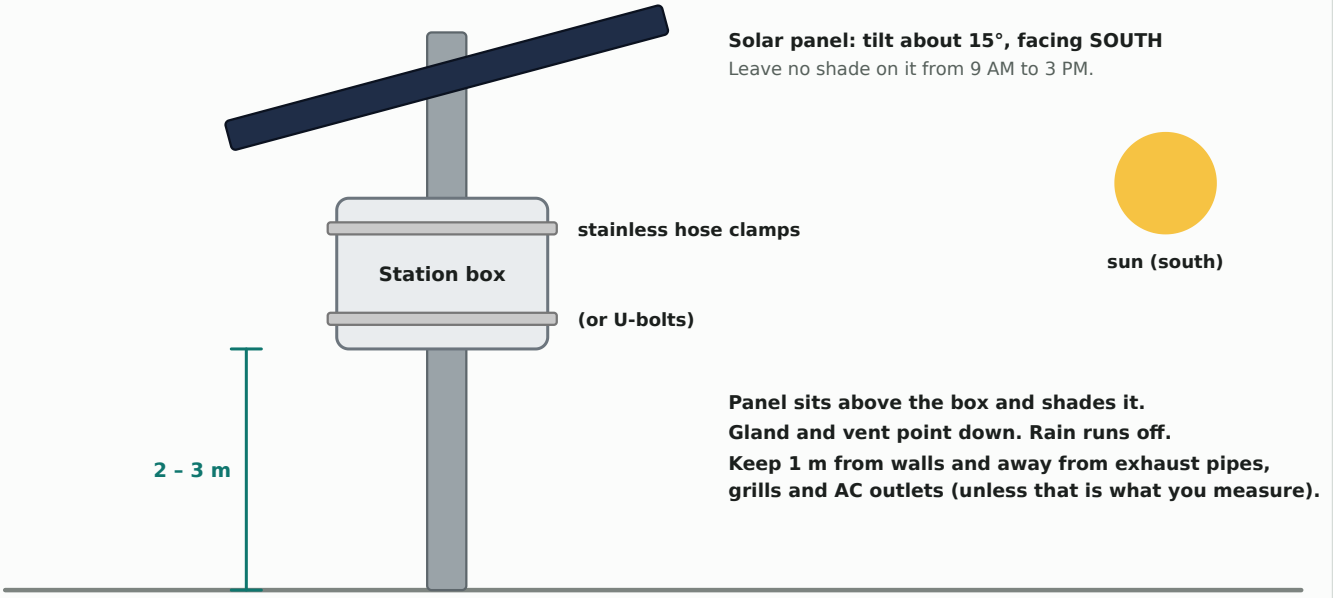



**Figure 15.** Mounting on a pole. The sensor should breathe the same air people do: 2 to 3 m above the ground, in open air.

1. Fix the box to the pole with two stainless hose clamps (or U-bolts) through its mounting ears or a backing plate.

2. Mount the solar panel above the box on a small bracket, tilted about 15°, facing south. It also shades the box.

3. Check that the gland and vent point down. Tighten everything once more.

4. Note the date, place and photos for the paper's deployment section.

STAGE 17 5 min

# See it in the app

The station adds itself. There is nothing to type in the app.

1. Open the Ginhawa app. A new card appears with **"Just added"** within a minute of the first upload.

2. After the first GPS fix (up to 15 minutes, outdoors) the station renames itself after its barangay. Admins can rename it in **Settings → Admin**.

3. Watch it for a day: PM values update every 5 minutes, VOC/NOx settle after about an hour, and the battery level stays up on sunny days.

REFERENCE

# Troubleshooting

Find the symptom, try the fixes in order.

| SYMPTOM | LIKELY CAUSE | FIX |
|---|---|---|
| Nothing turns on | Fuse out or blown, BMS asleep | Check the fuse with continuity. Put the panel in sun or touch the CN3791 BAT to wake the BMS. Measure VBAT bus. |
| Fuse blows at once | Short circuit or reversed wire | Cells out. Continuity between VBAT and GND bus must not beep. Look for solder bridges and reversed capacitors. |
| I²C scan finds nothing | SDA/SCL swapped, no 3.3 V, no common GND | Swap D21/D22 wires. Check 3V3 rail. Check ESP32 board GND links to the power board GND bus. |
| Only 0x40 found | SEN55 has no 5 V or wrong pin order | Measure 5 V between SEN55 cable pins 1 and 2. Recheck pin 1 with continuity. |
| Only 0x69 found | INA219 VCC or I²C wires | Check INA219 VCC = 3.3 V and its SDA/SCL wires. |
| Modem silent | TX/RX swapped, no VBAT, PWRKEY | Swap TXD/RXD. Measure VBAT at the modem. Press its power button if present. |
| Modem resets when connecting | Voltage dips during 4G bursts | Shorter, thicker VBAT/GND wires. Add the third 1000 µF capacitor. Charge the battery. |
| AT+CSQ = 99,99 | No signal or antenna on wrong plug | LTE antenna on MAIN. Try near a window or outside. Try another network SIM. |
| No GPS fix | Antenna cannot see the sky | Test outdoors for 5 minutes. GPS antenna flat side up, nothing metal above it. |
| "Connecting to Wi-Fi… failed" | Wrong name/password, 5 GHz network | Check config.h spelling. The ESP32 only joins 2.4 GHz Wi-Fi. |
| "Enrollment failed (HTTP 401/403)" | ENROLL_SECRET differs, or station blocked | Copy ENROLL_SECRET exactly from .env. Ask an admin to unblock the device. |
| "Upload failed (HTTP -1)" | Server not reachable | Laptop on, Docker running, same Wi-Fi, correct IP in API_BASE_URL. |
| Battery drops every day | Panel shaded or charger wiring | Check the CN3791 charging LED in sun. Clean the panel. Move it away from shade. |

REFERENCE

# Complete wiring list

Every connection in the station. Tick them off with the multimeter in continuity mode before the first full power-up.

| FROM | TO | WIRE | BUILT IN |
|---|---|---|---|
| Holder red lead | BMS B+ | 18 AWG | Stage 3 |
| Holder black lead | BMS B− | 18 AWG | Stage 3 |
| BMS P+ (or B+) | Fuse holder → INA219 VIN+ | 18 AWG | Stage 5 |
| INA219 VIN− | VBAT bus | 18 AWG | Stage 5 |
| BMS P− | GND bus | 18 AWG | Stage 5 |
| Solar panel + / − | CN3791 SOLAR + / − | panel cable | Stage 6 |
| CN3791 BAT + / − | VBAT bus / GND bus | 22 AWG | Stage 6 |
| TPS63020 VIN / GND | VBAT bus / GND bus | 22 AWG | Stage 7 |
| TPS63020 VOUT / GND | ESP32 board 3V3 rail / GND rail | 22 AWG | Stage 9 |
| MT3608 VIN+ / VIN− | VBAT bus / GND bus | 22 AWG | Stage 8 |
| MT3608 VOUT+ | SEN55 pin 1 (VDD) | 22 AWG | Stage 11 |
| MT3608 VOUT− | ESP32 board GND rail | 22 AWG | Stage 11 |
| ESP32 3V3 pin | 3V3 rail | under board | Stage 9 |
| ESP32 GND pin | GND rail | under board | Stage 9 |
| ESP32 board GND rail | Power board GND bus | 18 AWG | Stage 9 |
| R3 4.7 kΩ | D21 ↔ 3V3 rail | resistor | Stage 9 |
| R4 4.7 kΩ | D22 ↔ 3V3 rail | resistor | Stage 9 |
| SEN55 pin 2 (GND) | GND rail | GH cable | Stage 11 |
| SEN55 pin 3 (SDA) | ESP32 D21 | GH cable | Stage 11 |
| SEN55 pin 4 (SCL) | ESP32 D22 | GH cable | Stage 11 |
| SEN55 pin 5 (SEL) | GND rail | GH cable | Stage 11 |

| FROM | TO | WIRE | BUILT IN |
|---|---|---|---|
| SEN55 pin 6 | not connected | — | Stage 11 |
| INA219 VCC / GND | 3V3 rail / GND rail | 22 AWG | Stage 11 |
| INA219 SDA / SCL | ESP32 D21 / D22 | 22 AWG | Stage 11 |
| A7670SA VBAT / GND | VBAT bus / GND bus (+ 2 × 1000 µF) | 18 AWG | Stage 12 |
| A7670SA TXD | ESP32 RX2 (GPIO16) | 22 AWG | Stage 12 |
| A7670SA RXD | ESP32 TX2 (GPIO17) | 22 AWG | Stage 12 |
| A7670SA PWRKEY | ESP32 D4 (GPIO4) | 22 AWG | Stage 12 |
| A7670SA VCCIO (if present) | 3V3 rail | 22 AWG | Stage 12 |
| ESP32 VIN | nothing | — | — |

REFERENCE

# Parts and tools

What the team bought, and what is still needed before assembly.

## Bought

- [ ] SIMCom A7670SA LTE + GNSS breakout, with LTE and GPS antennas × 1
- [ ] ESP32 DevKit V1 (already owned) × 1
- [ ] Sensirion SEN55 (already owned) × 1
- [ ] Solar panel 6 V 6 W, 270 × 180 mm × 1
- [ ] CN3791 solar charger, 6 V version × 1
- [ ] Beston 18650 65N-26, 2600 mAh (3 used + 1 spare) × 4
- [ ] BMS 1S 3.7 V 10 A × 1
- [ ] 3-cell 18650 parallel holder × 1
- [ ] TPS63020 buck-boost module, 3.3 V × 1
- [ ] MT3608 boost module × 1
- [ ] INA219 module × 1
- [ ] Blade fuses (standard size), 5 A × set
- [ ] 16 AWG waterproof inline fuse holder, standard × 1
- [ ] 18 AWG silicone wire, red and black × 2 m each
- [ ] 22 AWG silicone wire, red and black × 1 m each
- [ ] JST GH 1.25 mm 6-pin double-ended cable × 5
- [ ] 4.7 kΩ resistors × 50
- [ ] 1000 µF 25 V capacitors × 20
- [ ] PG7 cable glands, black × 5
- [ ] M12 × 1.5 breather vent, black × 5
- [ ] Perfboard 7 × 9 cm (2) and 5 × 7 cm (1) × 3
- [ ] Female pin headers 1 × 40 × 10

## Still needed

- [ ] IP65 ABS junction box, about 150 × 100 × 70 mm to buy
- [ ] Nano-SIM (DITO, Smart or Globe) with data load to buy
- [ ] Stainless hose clamps or U-bolts, small panel bracket to buy
- [ ] 2 mm Allen key, 12 mm and 3 mm drill bits to buy or borrow
- [ ] Double-sided foam tape, zip ties, electrical tape to buy
- [ ] Soldering iron, solder, flux, side cutter, wire stripper borrow
- [ ] Multimeter borrow (required)
- [ ] USB data cable for the ESP32 probably owned
- [ ] Hot glue or epoxy, small file, marker borrow

- [ ] M3 × 10 + 6 nylon standoffs × 50
- [ ] M3 nuts × 10+
- [ ] M3 × 8 button screws × 50
- [ ] Heat-shrink tubing kit × 164 pcs
- [ ] Silica gel packs × 1 pack
- [ ] Stainless insect mesh × 1 piece

Ginhawa capstone project. Pin assignments match firmware/ginhawa_node (SDA D21, SCL D22, modem RX2/TX2 GPIO16/17, PWRKEY GPIO4) and docs/diagrams/03-circuit-schematic.